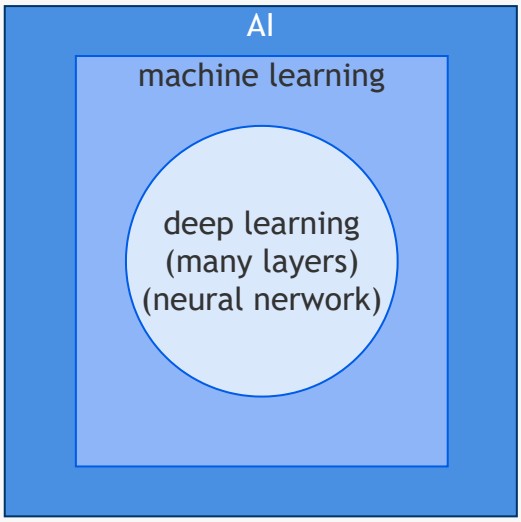
AI outline



Definition: Generative AI

讓computer學會產生複雜而有結構的物件

- 複雜 : 無限可能
- 有結構 : 由有限的基本單位(toekn)所構成

Retrieval Augmented Generation(RAG)

Context Engineering的核心目標

- 避免塞爆Context(把需要的放進去,不需要的清出來)
- 常用招數
  - Select
    - 挑選需要的內容, e.g. RAG, Tool RAG, Memory RAG
  - Compress
  - Multi-Agent

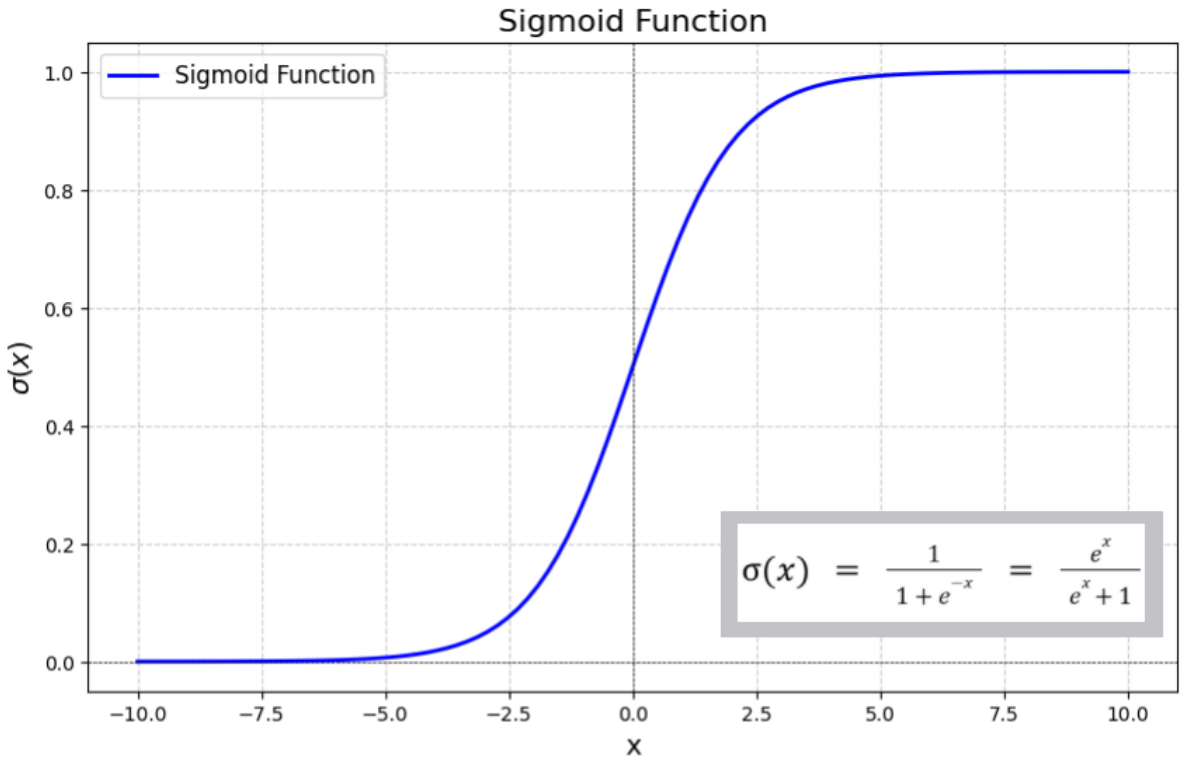
Machine Learning

1. prepare data

- split to train data & test data

2. set a model

- Simple Linear Regression
  - $y = \beta_0 + \beta_1 x + \epsilon$
    - 其中 β_0 為截距，β_1 為斜率，ϵ 為隨機誤差。
- Multiple Linear Regression
  - $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$
- Logistic Regression(分類問題/classification)
  - $P(y=1|x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\cdots+\beta_n x_n)}}$



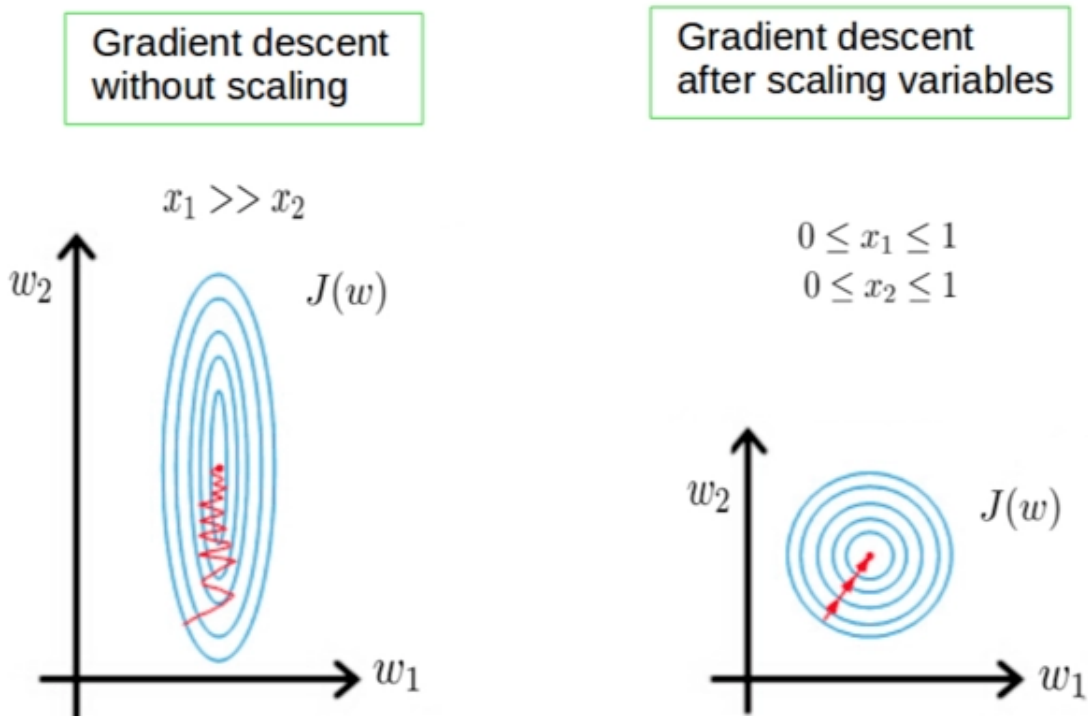
3. set Cost function/Loss function

- Mean Squared Error, MSE
  - $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
    - n : 訓練資料的總數 (Train Data Size / Batch Size).
    - y_i : 第 i 筆資料的真實標籤 (Ground Truth).
    - $\hat{y}_i$: 模型對第 i 筆資料的預測值 (Prediction)。

4. set optimizer

- gradient descent
  - $w_{new} = w_{old} - \eta \cdot \nabla L(w)$
    - $\nabla L(w)$ 代表損失函數 L 在當前參數 w 位置的「斜率」或「坡度」
    - η : The Learning Rate
  - accelerate gradient descent

- Feature Scaling



5.Train the model

- **Use gradient descent to train the model, accelerating convergence to the minimum loss and yielding the optimal model(Find the best $\beta_0, \beta_1, \beta_2, \beta_3, \ldots, \epsilon$).**

6. use test data to test your model